
DATABASE HISTOGRAMS

E0 261

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MOTIVATION

- System R's assumption of **uniform** distribution of values over data domain rarely holds true in practice
⇒ Garbage In, Garbage Out



Solutions

- Use a “classical” distribution (e.g. Gaussian, Exponential, Zipf, etc.) to model the data
 - Problem is that real-life data is often not like these also !
 - Further, not all distributions are easily computable !
- Approximate the distribution using **histograms**
 - Several flavors available
 - Maintenance could be a pain
- Use sampling for dynamic estimation
 - Expensive since done at run-time
 - Space-efficient

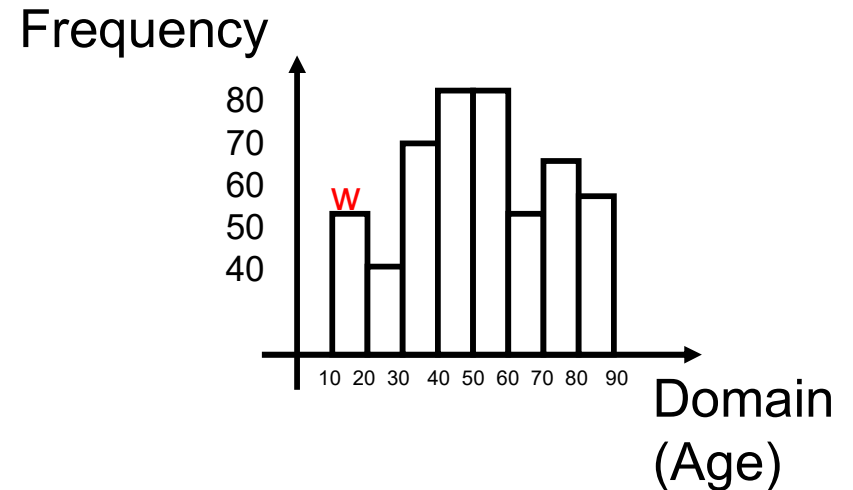


HISTOGRAMS

- Domain of attribute **A** is partitioned into **B** buckets, and a **uniform** distribution is assumed within each bucket. That is, frequency of a value in a bucket is approximated by the average of the frequencies of all domain values assigned to the bucket.
- Trivial histogram: Single bucket that assigns the same frequency ($N \div V$) to all attribute values.
 - equivalent to System R approach

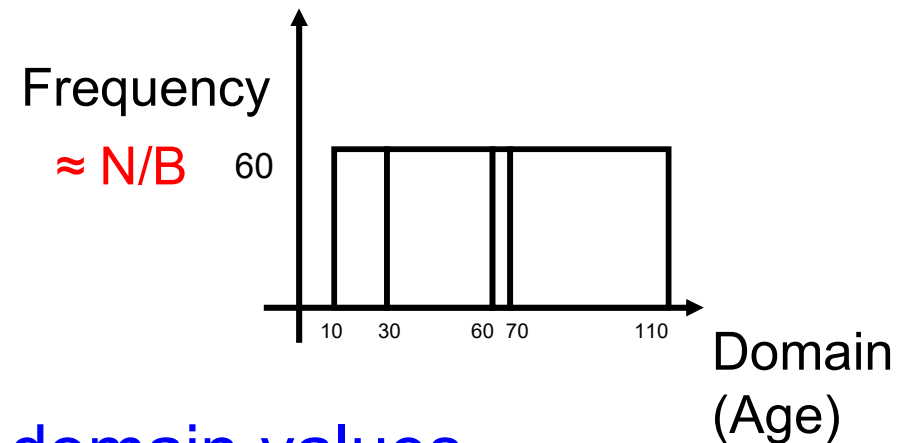


EQUI-WIDTH HISTOGRAMS



- Frequency versus ordered domain values
- All buckets of same width w
- $\text{MaxErr} [\text{Sel } <X] = \text{FreqFraction} [X_{\text{PB}}]$
- Easy to construct and maintain
- Example: CRICKET !

EQUI-DEPTH HISTOGRAMS



- Frequency versus ordered domain values
- All buckets of same height (or depth)
- $\text{MaxErr} [\text{Sel} < X] = 1 / B$
- Comparatively difficult to construct and maintain
 - Require to sort the relation first to find the B “quantiles”
 - Or use index if available
 - Cheaper approximate technique based on sampling typically used

Sampling Technique

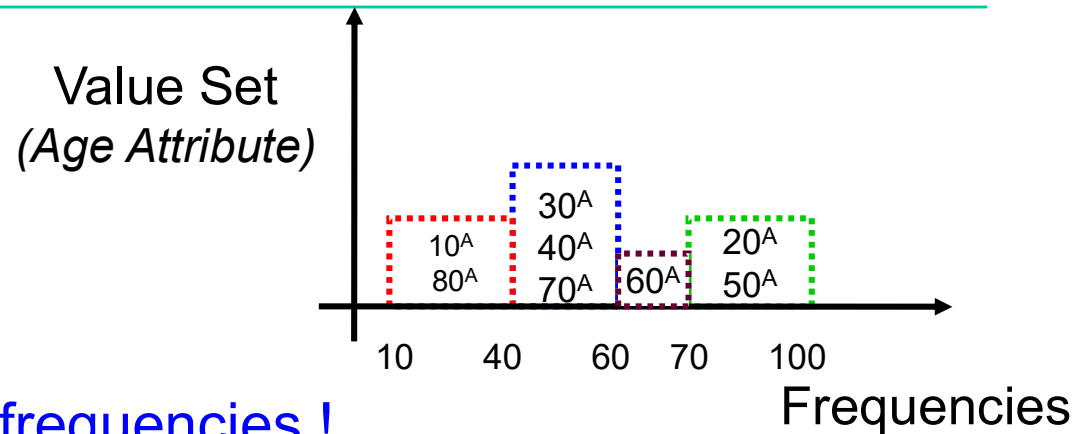
- Take a random sample of database, sort the sampled tuples, and use boundaries established by them to form the approximate buckets
- How many samples to take ?
 - Based on **Kolmogorov** statistic



Kolmogorov Statistic

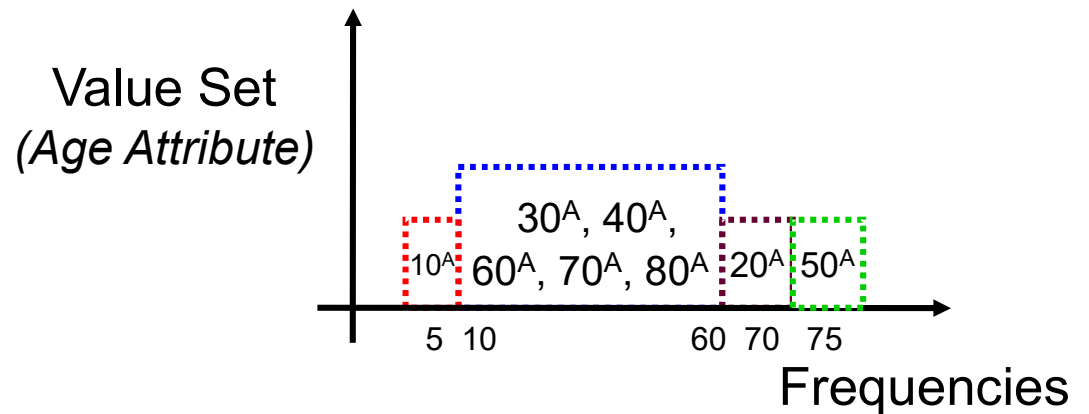
- Let α be the fraction of tuples that satisfy a property (in our case, property is query box).
- Let β be the fraction of tuples in sample that lie in same box. Then, the K-statistic is that $|\alpha - \beta| \leq d$ (precision) with probability $\geq p$ (confidence) if sample size is $\geq n$.
- Given p and d , n can be evaluated. For example, if $d = 0.05$ and $p = 0.99$, $n = 1064$
 - Note, does not depend on number of tuples in database!

SERIAL HISTOGRAMS



- Domain values versus ordered frequencies !
- Frequency assigned to domain values in a bucket is avg frequency of bucket or middle frequency of bucket depending on information stored.
- **Optimal** histogram is a serial histogram
 - paper covered in TIDS (E0 361) course
 - in particular, the serial histogram that minimizes $\sum n_i V_i$, where n_i is number of attribute values placed in bucket B_i , and V_i is variance of these frequencies
- Difficult to construct and maintain
 - Queries are usually on value domain, not frequency domain
 - Have to explicitly represent all domain values – ranges not possible
 - Identification of optimal is exponential in number of buckets

END-BIASED HISTOGRAMS



- Special case of serial histograms
- Highest / lowest sets of frequencies maintained in separate individual buckets
- Remaining (middle) frequencies are approximated together in a single bucket
- **Optimal** end-biased histogram minimizes $\sum n_i V_i$
- Performance “close” to that of optimal serial histogram
- Comparatively easy to construct and maintain
 - Attribute Values corresponding to the “middle” bucket do not have to be stored explicitly: can be identified by negation.
 - **Linear time complexity** in number of buckets to identify optimal

Serial Histogram Construction

- Data Collection:

```
select A, Count(*) as Frequency  
from R  
group by A  
order by Frequency
```

 - requires sorting
- Approximate by sampling when building high-end-biased serial histograms



In Practice

- Histograms used in most commercial systems.
- Approximate equi-depth histograms built on individual attributes with about 10-20 buckets using a one-pass sampling (usually 1000-2000 samples) approach.
- Results in large errors for multi-attribute queries, but technique proposed in SIGMOD 99 for efficiently constructing and maintaining multi-dimensional histograms (covered in TIDS course)

END DATABASE HISTOGRAMS

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